Topology Exam 1 Cheatsheet

Raymond Langehennig

March 2, 2026

Topological Spaces

topology A set T of subsets of X such that

1. $X \in T$
2. $\varnothing \in T$
3. T is closed under *arbitrary unions* and *finite intersections*

basis A collection $\mathcal{B}$ of subsets of X such that

1. $\forall x \in X, \exists B \in \mathcal{B} : x \in B$
2. $\forall B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2 \implies \exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The set $T_{\mathcal{B}} = \{\bigcup \mathcal{F} \mid \mathcal{F} \subset \mathcal{B}\}$

fineness

T is finer than $S \iff S \subset T$

subspace (Y, T_Y) where $Y \subset X$ and $T_Y = \{Y \cap U \mid U \in T\}$

closure $\overline{A} = \mathrm{Cl}(A) = \bigcap \{U \mid \complement_X U \in T \wedge A \subset U\}$ is the smallest closed set containing A

interior $\mathrm{Int}(A) = \bigcup \{U \mid U \in T \wedge U \subset A\}$ is the largest open set contained in A

Special cases

discrete $T = \mathcal{P}(X)$

Hausdorff

metric For a metric space,

subspace

On $\mathbb{R}$:

standard On $\mathbb{R}^n$, generated by $\{B_r(p) \mid r \in \mathbb{R} \wedge p \in \mathbb{R}^n\}$

order

upper

k-topology